

OpenMed Instruments (fictional)

SP-100 Portable Suction Pump

SERVICE MANUAL

Document SP100-SM-EN · Revision 2.1

DEMO MANUAL. This is an original, fictional document created to demonstrate the Taller app. The device does not exist. Do not use for real maintenance. For real work, import the manufacturer's manual of your actual equipment.

Contents

Section	Page
1. Safety warnings	3
2. Technical specifications	4
3. Device overview	5
4. Principle of operation	6
5. Error codes	7
6. Troubleshooting — suction problems	8
7. Troubleshooting — power and motor	9
8. Troubleshooting — overheating and noise	10
9. Preventive maintenance schedule	11
10. Cleaning and disinfection	12
11. Spare parts list	13
12. Vacuum test and calibration	14

1. Safety warnings

WARNING — Electrical hazard. Disconnect the mains plug before opening the housing. Capacitor C4 can hold charge for up to 60 seconds after power-off. Wait one minute before touching the motor driver board.

WARNING — Biohazard. The collection jar, lid, float valve and patient tubing must be treated as contaminated. Wear gloves. Disinfect before any service intervention (see section 10).

CAUTION. Never run the pump without the bacterial filter installed. Fluid ingress into the pump head is the most common cause of premature diaphragm failure.

Only trained biomedical technicians should service this device. After any repair, perform the vacuum test of section 12 before returning the unit to clinical use. Never bypass the overflow protection float or the thermal cut-out switch.

2. Technical specifications

Parameter	Value
Mains supply	100–240 V AC, 50/60 Hz, 90 W
Internal battery	12 V 2.6 Ah sealed lead-acid, 45 min autonomy
Maximum vacuum	–80 kPa (–600 mmHg) $\pm$ 5 %
Free air flow	30 L/min minimum at pump inlet
Vacuum regulator range	–10 to –80 kPa, continuous
Collection jar	1.0 L polycarbonate, autoclavable at 121 °C
Bacterial filter	Hydrophobic 0.2 μ m, single use, part FT-020
Fuse	T2AL 250 V, 5×20 mm, two units on mains inlet
Pump type	Oil-free double-diaphragm, brushless DC motor
Duty cycle	Continuous (thermal cut-out at 70 °C)
Weight / dimensions	4.8 kg · 330 × 210 × 240 mm
Noise level	< 55 dB(A) at 1 m

3. Device overview

Main assemblies, from the patient side to the mains inlet:

Ref	Assembly	Notes
A1	Collection jar with float valve	Float closes suction when jar is full (overflow protection)
A2	Bacterial filter FT-020	Between jar lid and pump inlet; replace if wet or discoloured
A3	Vacuum regulator and gauge	Rotary valve; gauge marked in kPa and mmHg
A4	Pump head (double diaphragm)	Two valve plates, two diaphragms; kit PK-110
A5	BLDC motor and driver board	Hall-sensor commutated; connector J3, 6-pin
A6	Main board (PSU + control)	Fuses F1/F2, capacitor C4, error LED display
A7	Battery pack	Connector J5; replace every 24 months
A8	Mains inlet module	IEC C14 with switch and fuse drawer

The housing opens with four Torx T15 screws under the rubber feet. The pump head is mounted on two anti-vibration studs; do not overtighten (0.8 N·m).

4. Principle of operation

The brushless motor drives an eccentric that flexes two diaphragms in opposition. Reed valves on the valve plates rectify the airflow, producing vacuum at the inlet port. The control board regulates motor speed to hold the vacuum selected on the regulator, monitors motor current, battery voltage and pump-head temperature, and reports faults as error codes E1–E8 on the front panel display.

When mains power is present the battery charges at 0.4 A; the unit switches to battery automatically on mains failure. If jar overflow occurs, the float valve seals the lid and vacuum rises sharply while flow stops — this protects the filter and pump, and typically triggers error E5 (blocked inlet).

Typical current draw at –60 kPa is 1.1–1.4 A from the 12 V rail. A reading above 2.0 A with low vacuum indicates a mechanical fault in the pump head (see section 6).

5. Error codes

Codes appear on the front display. Press and hold MODE for 3 s to show the last five stored codes with hour-meter stamps.

Code	Meaning	First actions
E1	Mains fuse fault / no mains detected	Check fuses F1 and F2 (T2AL 250 V) in the inlet drawer; check mains cable and socket
E2	Battery low (< 10.8 V)	Connect mains 8 h; if E2 returns quickly, test battery capacity, replace pack (24 months)
E3	Motor stall / overcurrent	Disconnect pump head cam; if motor spins free, inspect diaphragms and valve plates
E4	Pump-head over-temperature (> 70 °C)	Let unit cool 30 min; clean air vents; check fan spins; verify duty conditions (ambient < 35 °C)
E5	Blocked inlet / no flow with high vacuum	Empty jar (float may be sealed); replace bacterial filter FT-020 if wet; check tubing for blockages
E6	Vacuum sensor out of range	Check sensor hose to main board port P1 for cracks; re-zero per section 12; replace sensor if needed
E7	Hall sensor / commutation fault	Re-seat connector J3; measure 5 V at J3 pin 1; replace motor if any Hall line stays high
E8	Control board internal fault	Power-cycle once; if E8 persists, replace main board MB-100 — do not attempt component repair

6. Troubleshooting — suction problems

6.1 No suction at the patient tube

Work from the patient side towards the pump: (1) Jar lid not seated or gasket damaged — refit lid, inspect gasket G-101 for cuts; (2) Float valve stuck closed after overflow — empty jar, rinse float, verify free movement; (3) Bacterial filter FT-020 wet or clogged — replace, never wash; (4) Tubing kinked or connectors cracked — replace tubing set TS-210; (5) Regulator fully open to atmosphere — rotate clockwise to increase vacuum; (6) If still no suction with the inlet blocked by a finger and gauge at zero: pump head fault — see 6.3.

6.2 Weak suction (vacuum below -40 kPa with inlet blocked)

Most common causes in field units: leaking jar gasket (replace G-101), perished internal silicone hose between filter port and pump inlet (replace, 6×9 mm silicone), and worn diaphragms after ~3,000 h (fit pump kit PK-110). Soapy water on joints shows leaks as bubbles while running. Verify the gauge itself against a known-good gauge before replacing parts.

6.3 Pump head service

Remove four M4 screws on the head cover. Mark diaphragm orientation before removal. Inspect reed valves for corrosion or debris; clean seats with isopropyl alcohol only. Fit new diaphragms with the kit's new screws, torque 0.8 N·m in a cross pattern. Run 5 min and perform the section 12 vacuum test.

7. Troubleshooting — power and motor

7.1 Unit completely dead

Check in order: mains socket live (test lamp); mains cable continuity; fuses F1/F2 in the inlet drawer — replace only with T2AL 250 V; inlet switch continuity; 12 V present at battery connector J5 (battery may be deeply discharged — unit can run from mains with battery disconnected for diagnosis). A blown fuse that blows again immediately indicates a shorted rectifier on the main board or a seized motor — do not keep replacing fuses.

7.2 Runs on mains but not on battery

Measure battery open-circuit voltage: below 11.5 V after 8 h charging means the pack is worn — replace (part BT-126). Check the 3 A blade fuse on the battery lead. If charging LED never lights, measure 13.8 V at J5 with mains on; absent → charger section fault on main board.

7.3 Motor tries to start then stops (often with E3)

Fluid ingress gums the pump head: remove head cover and check for liquid. Turn the eccentric by hand — it must rotate smoothly. Verify J3 connector fully latched (vibration loosens it). Measure motor phase resistance: 0.8–1.2 Ω between any two phase pins; open or shorted windings → replace motor MT-100.

8. Troubleshooting — overheating and noise

8.1 Overheating / repeated E4

Clean dust from the vent grilles and fan with a dry brush. Confirm the fan runs above -30 kPa load. Check ambient temperature and that the unit is not enclosed in a cabinet. A pump head rebuilt without the anti-vibration studs, or run continuously at maximum vacuum in ambient above 40 °C, will cycle on the thermal cut-out — this is protection, not a fault.

8.2 Excessive noise or vibration

Rattling: loose housing screws or cracked anti-vibration studs (replace in pairs, part AV-014). Knocking at low vacuum: worn eccentric bearing — replace with pump kit PK-110. High-pitched whine: reed valve chatter, usually debris on a valve seat — clean per 6.3. After any noise repair, run at -60 kPa for 10 minutes and confirm temperature stays below 60 °C at the head.

8.3 Display dead but pump runs

Re-seat display ribbon cable at connector J7. If segments are missing, replace display module DP-100; the pump remains safe to use short-term — vacuum is still regulated mechanically.

9. Preventive maintenance schedule

Interval	Task	Parts
Daily (user)	Empty and disinfect jar; check filter dry; test suction with finger over inlet	
Monthly	Inspect tubing and lid gasket; clean vents; test battery runtime ≥ 30 min if damaged	30-101
6 months	Vacuum test (section 12); check fuse drawer; verify float valve free—	
12 months	Replace bacterial filter stock rotation; battery capacity test; electrical safety test (Class II leakage limits)	ET-520
24 months	Replace battery pack; replace jar gasket set; consider diaphragm BT-126, 0001, PK-110	BT-126, 0001, PK-110

Record every intervention in your maintenance log with date, hour-meter reading and parts used. Units with complete logs show roughly half the unplanned downtime of unlogged units in our field data.

10. Cleaning and disinfection

Jar, lid and float: wash with warm water and detergent, then disinfect (chlorine 0.5 % for 10 min or autoclave 121 °C / 15 min — polycarbonate jar only, check for crazing after repeated cycles). Tubing sets are single-patient use where supplies allow; otherwise disinfect chemically, never autoclave PVC tubing.

Housing: wipe with 70 % isopropyl alcohol. Never spray liquids at the vents. The bacterial filter FT-020 cannot be cleaned or reused: a wet filter both blocks flow (E5) and loses its barrier function.

Before opening the unit for service, run 30 s with a jar of disinfectant connected, wipe external surfaces, and wear gloves throughout.

11. Spare parts list

Part no.	Description	Typical replacement
FT-020	Bacterial filter 0.2 µm hydrophobic	When wet/discoloured; 12-month stock rotation
G-101	Jar lid gasket, silicone	On damage; every 24 months
TS-210	Tubing set, patient side, 2 m	On damage or per hygiene policy
PK-110	Pump kit: 2 diaphragms, 2 valve plates, screws	~3,000 h or on fluid ingress
BT-126	Battery 12 V 2.6 Ah SLA	24 months
MT-100	BLDC motor with Hall sensors	On winding/Hall failure
MB-100	Main board (PSU + control)	On E8 persistent
DP-100	Display module	On display failure
AV-014	Anti-vibration studs (pair)	On cracking/noise
FU-T2A	Fuse T2AL 250 V 5×20 mm (×10)	As needed — find root cause of repeat failures

Generic equivalents: the fuse, silicone hose (6×9 mm), and SLA battery are standard commodity parts available in most markets; match ratings exactly.

12. Vacuum test and calibration

12.1 Performance test (after every repair)

(1) Fit clean jar, dry filter, close inlet with the test cap. (2) Run at maximum: gauge must reach -75 kPa or better within 10 s. (3) Stop pump: vacuum must not fall more than 5 kPa in 60 s (leak test). (4) Open regulator to -30 kPa: reading must hold ± 3 kPa for 5 min. (5) Free-flow: with inlet open, a 1 L jar fills with water lifted 0.5 m in under 25 s (≈ 30 L/min air equivalent).

12.2 Vacuum sensor re-zero

With the unit off and inlet open to atmosphere, hold MODE + POWER for 5 s until 'CAL' shows, then press MODE once. The display shows '0.0'. If E6 persists after re-zero, replace sensor S2 on the main board (plug-in part, no soldering).

12.3 Electrical safety

After any opening of the housing: Class II touch-current test per your local standard (IEC 62353 limits). Record results in the maintenance log.